

n-Propyl gallate repeat: simple layout

one row per culture • one dose per column • bottom row all controls • 24 vials per plate

15 °C

fill left to right, row by row

	1	2	3	4	5	6
culture 1 A	A1	A2	A3	A4	A5	A6
	0.05 mM	0.1 mM	0.25 mM	0.5 mM	1.0 mM	2.0 mM
culture 2 B	B1	B2	B3	B4	B5	B6
	0.05 mM	0.1 mM	0.25 mM	0.5 mM	1.0 mM	2.0 mM
culture 3 C	C1	C2	C3	C4	C5	C6
	0.05 mM	0.1 mM	0.25 mM	0.5 mM	1.0 mM	2.0 mM
controls D	D1	D2	D3	D4	D5	D6
	control	control	control	control	control	control
	C1	C2	C3	C1	C2	C3

dose increases left → right

27 °C

dose order REVERSED

	1	2	3	4	5	6
culture 1 A	A1	A2	A3	A4	A5	A6
	2.0 mM	1.0 mM	0.5 mM	0.25 mM	0.1 mM	0.05 mM
culture 2 B	B1	B2	B3	B4	B5	B6
	2.0 mM	1.0 mM	0.5 mM	0.25 mM	0.1 mM	0.05 mM
culture 3 C	C1	C2	C3	C4	C5	C6
	2.0 mM	1.0 mM	0.5 mM	0.25 mM	0.1 mM	0.05 mM
controls D	D1	D2	D3	D4	D5	D6
	control	control	control	control	control	control
	C1	C2	C3	C1	C2	C3

dose increases right → left

Legend

 solvent control (n=6)	 0.05 mM	 0.1 mM	 0.25 mM
 0.5 mM	 1.0 mM	 2.0 mM	

The dose order is reversed on the 27 °C plate. Filling is just as simple, but if the plate has a left-right gradient it now pushes the two temperatures in opposite directions, so it cannot fake the same dose response at both. That is the one safeguard kept.

nPG repeat — well by well, 15 °C plate

Every vial: your usual culture volume + 20 µL from the stock column. Solvent identical in all 24.

well	culture	20 µL slot	final	note
 A1	culture 1	nPG stock 12.5 mM	0.05 mM	also in first run
 A2	culture 1	nPG stock 25 mM	0.1 mM	also in first run
 A3	culture 1	nPG stock 62.5 mM	0.25 mM	NEW dose
 A4	culture 1	nPG stock 125 mM	0.5 mM	NEW dose
 A5	culture 1	nPG stock 250 mM	1.0 mM	NEW dose
 A6	culture 1	nPG stock 500 mM	2.0 mM	NEW dose
 B1	culture 2	nPG stock 12.5 mM	0.05 mM	also in first run
 B2	culture 2	nPG stock 25 mM	0.1 mM	also in first run
 B3	culture 2	nPG stock 62.5 mM	0.25 mM	NEW dose
 B4	culture 2	nPG stock 125 mM	0.5 mM	NEW dose
 B5	culture 2	nPG stock 250 mM	1.0 mM	NEW dose
 B6	culture 2	nPG stock 500 mM	2.0 mM	NEW dose
 C1	culture 3	nPG stock 12.5 mM	0.05 mM	also in first run
 C2	culture 3	nPG stock 25 mM	0.1 mM	also in first run
 C3	culture 3	nPG stock 62.5 mM	0.25 mM	NEW dose
 C4	culture 3	nPG stock 125 mM	0.5 mM	NEW dose
 C5	culture 3	nPG stock 250 mM	1.0 mM	NEW dose
 C6	culture 3	nPG stock 500 mM	2.0 mM	NEW dose
 D1	culture 1	ethanol (no compound)	0 mM	
 D2	culture 2	ethanol (no compound)	0 mM	
 D3	culture 3	ethanol (no compound)	0 mM	
 D4	culture 1	ethanol (no compound)	0 mM	
 D5	culture 2	ethanol (no compound)	0 mM	
 D6	culture 3	ethanol (no compound)	0 mM	

nPG repeat — well by well, 27 °C plate

Every vial: your usual culture volume + 20 µL from the stock column. Solvent identical in all 24.

well	culture	20 µL slot	final	note
 A1	culture 1	nPG stock 500 mM	2.0 mM	NEW dose
 A2	culture 1	nPG stock 250 mM	1.0 mM	NEW dose
 A3	culture 1	nPG stock 125 mM	0.5 mM	NEW dose
 A4	culture 1	nPG stock 62.5 mM	0.25 mM	NEW dose
 A5	culture 1	nPG stock 25 mM	0.1 mM	also in first run
 A6	culture 1	nPG stock 12.5 mM	0.05 mM	also in first run
 B1	culture 2	nPG stock 500 mM	2.0 mM	NEW dose
 B2	culture 2	nPG stock 250 mM	1.0 mM	NEW dose
 B3	culture 2	nPG stock 125 mM	0.5 mM	NEW dose
 B4	culture 2	nPG stock 62.5 mM	0.25 mM	NEW dose
 B5	culture 2	nPG stock 25 mM	0.1 mM	also in first run
 B6	culture 2	nPG stock 12.5 mM	0.05 mM	also in first run
 C1	culture 3	nPG stock 500 mM	2.0 mM	NEW dose
 C2	culture 3	nPG stock 250 mM	1.0 mM	NEW dose
 C3	culture 3	nPG stock 125 mM	0.5 mM	NEW dose
 C4	culture 3	nPG stock 62.5 mM	0.25 mM	NEW dose
 C5	culture 3	nPG stock 25 mM	0.1 mM	also in first run
 C6	culture 3	nPG stock 12.5 mM	0.05 mM	also in first run
 D1	culture 1	ethanol (no compound)	0 mM	
 D2	culture 2	ethanol (no compound)	0 mM	
 D3	culture 3	ethanol (no compound)	0 mM	
 D4	culture 1	ethanol (no compound)	0 mM	
 D5	culture 2	ethanol (no compound)	0 mM	
 D6	culture 3	ethanol (no compound)	0 mM	

nPG repeat — stocks and notes

Ethanol stocks

n-propyl gallate, M = 212.20 g/mol. Serial dilutions from a 500 mM master.

500 mM	106.1 mg in 1.00 mL absolute ethanol	→ 2.0 mM ** check it dissolves **
250 mM	500 µL of 500 mM + 500 µL ethanol	→ 1.0 mM
125 mM	500 µL of 250 mM + 500 µL ethanol	→ 0.5 mM
62.5 mM	500 µL of 125 mM + 500 µL ethanol	→ 0.25 mM
25 mM	400 µL of 62.5 mM + 600 µL ethanol	→ 0.1 mM
12.5 mM	500 µL of 25 mM + 500 µL ethanol	→ 0.05 mM

At most 120 µL of any stock across both plates. Make 1 mL of each.

If 500 mM will not dissolve, drop the top dose to 1.0 mM and make those six wells extra controls.
Do not raise the solvent volume to compensate.

Layout

- Row A = culture 1, row B = culture 2, row C = culture 3. Row D = six solvent controls.
- One dose per column. Fill left to right, row by row, using the same tip per column.
- 15 °C plate runs 0.05 → 2.0 mM left to right. The 27 °C plate runs 2.0 → 0.05, i.e. reversed.
- Top dose 2.0 mM is 20x your previous maximum. Range spans 1.6 decades against 0.90 before.
- 0.05 and 0.1 mM carry over from the first run, so the two datasets overlap.

At the bench

- C1, C2, C3 must be three separately grown cultures, not one suspension split three ways.
- Every vial gets 20 µL: nPG stock where a dose is listed, plain ethanol in row D.
- Record which physical plate went to which incubator, and the start time for each.
- Select fit windows blind to dose, or use an automated rule applied to every curve.

Two things this layout cannot protect against

1. Dose is still assigned to plate column, so a left-right gradient enters the dose axis. Reversing the order on the 27 °C plate means it cannot produce the same apparent effect at both temperatures, which is the comparison you care about, but it does not remove the gradient.
2. There are no cell-free wells. If the dose series shows an effect you will not be able to say whether nPG reduced respiration or consumed oxygen itself. If you get a positive result, run three cell-free vials afterwards (medium + nPG 2.0 mM, 0.1 mM, ethanol) before interpreting it.